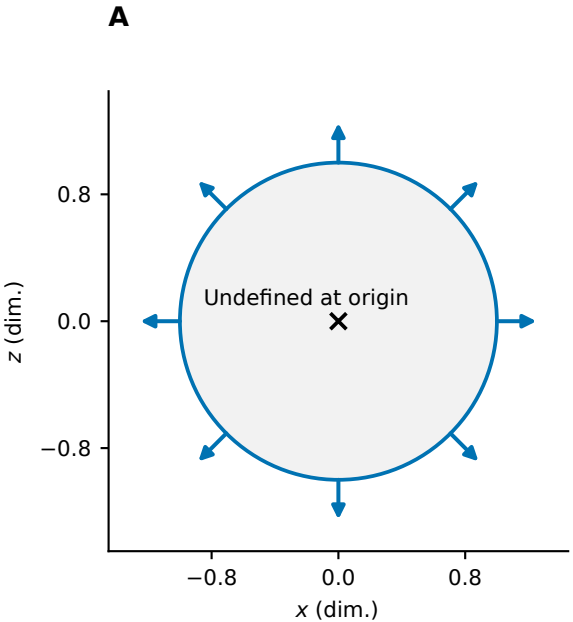
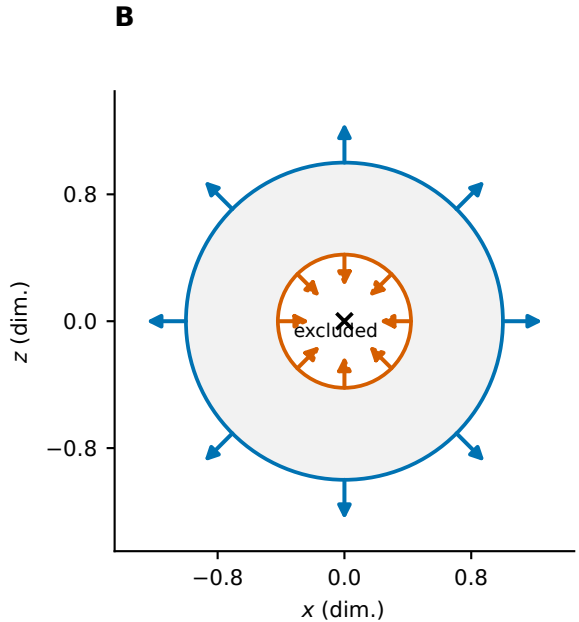


Ball: singularity invalidates Gauss



Shell: $\Phi_{out} = 4\pi, \quad \Phi_{in} = -4\pi$